

OpenClaw AI Weather Station Tutorial

Before starting, please complete the following from the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw Jetson-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solutions Configuration Test]. If you are currently only using TUI / WhatsApp, you can skip `04`. This article assumes the basic environment is already prepared and only covers the AI Weather Station project itself.

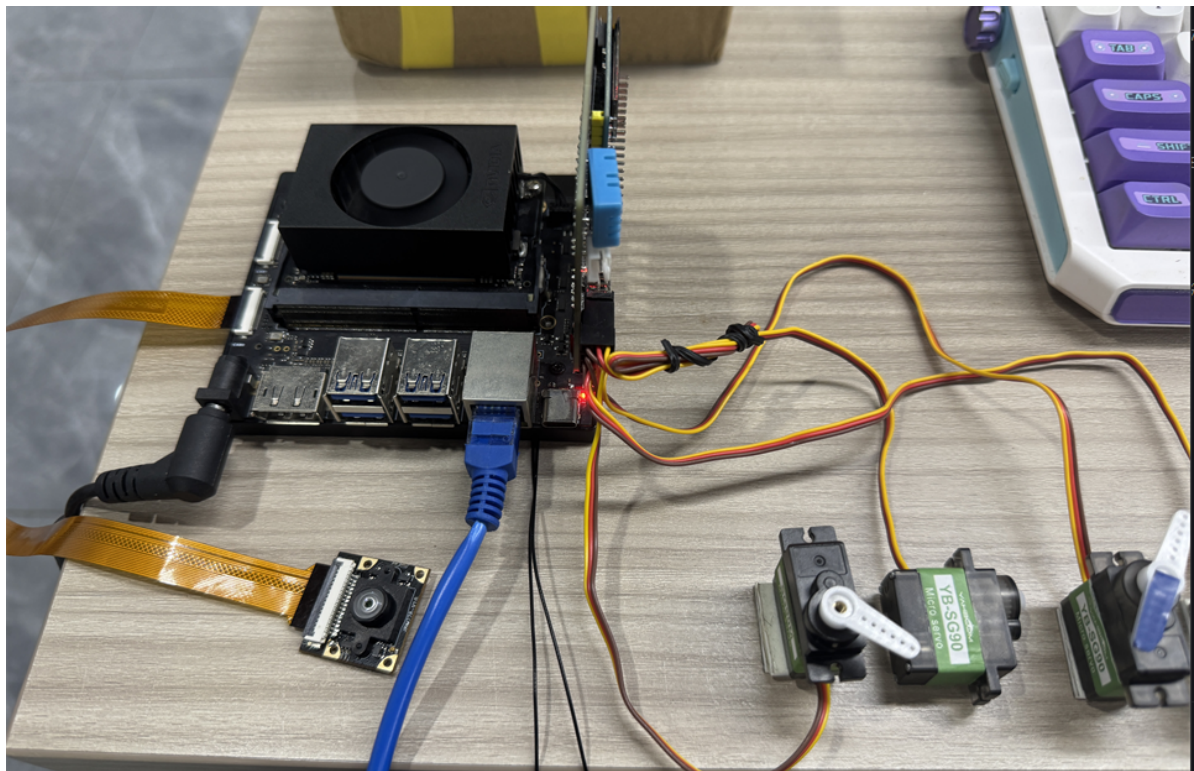
1. Tutorial Objectives

This article demonstrates OpenClaw's capabilities in a desktop weather station scenario through temperature/humidity collection, information display, and AI interpretation. After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, and RGB to build a local AI weather station
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: environment collection, AI broadcast, weather card, and status alert

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB LED module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

This tutorial focuses on a "local environment weather station", i.e., using DHT-based indoor temperature and humidity data for display and explanation.

4. Testing Project Basic Capabilities

Before starting this project, please complete a basic dialogue self-check using `openclaw tui`; if you haven't done so yet, see [05-Three Dialogue Solutions Configuration Test] first. After passing, proceed to the AI Weather Station specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
  
session agent:main:main  
connected | idle  
agent main | session main (openclaw-tui) | baillan/qwen3.5-plus | tokens 0/1.0m (0%)  
  
|  
  

```

Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display weather card on OLED
- If the environment is normal, turn on green light
- Give me a brief environment broadcast

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote environment viewing, classroom demonstrations	Suitable for remotely asking "how is the environment right now"	Demonstrate "remote weather station assistant"
TUI	Local debugging, threshold verification	Easiest for troubleshooting sensor and display logic	First run through the collection pipeline
Voice code	Voice broadcast, desktop assistant	Closer to daily usage experience	Use as the final demonstration solution

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to read aloud OpenClaw's reply content. You can run the following program in the terminal. The AI voice module dialogue does not need to run this program repeatedly.

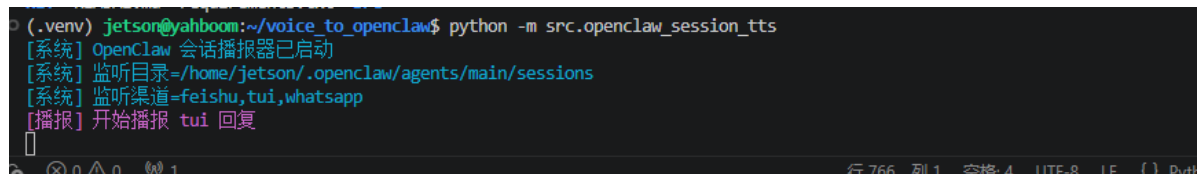
- Enable the broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether audio readback is enabled.  
After modification, save and run the following program in the terminal
```

- Run the following in the terminal:

```
cd ~/voice_to_openclaw  
source .venv/bin/activate  
python -m src.openclaw_session_tts
```



```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts  
[系统] OpenClaw 会话播报器已启动  
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions  
[系统] 监听渠道=feishu,tui,whatsapp  
[播报] 开始播报 tui 回复  
□
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Solutions Configuration Test]. After creation, you can remotely view environment status in WhatsApp, for example:

- How's the indoor environment now? If it's too hot, remind me

20:34



< 2



+86 153 3885 7526 (你)

给自己发消息

...waiting for me...

20:33 ✓✓

How is the indoor environment now? If it is too hot, remind me.

20:34 ✓✓

Current indoor environment readings:



Temperature: 29.7°C (85.5°F)



Humidity: 54%

Assessment: The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

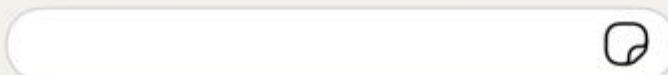
Recommendation:

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓



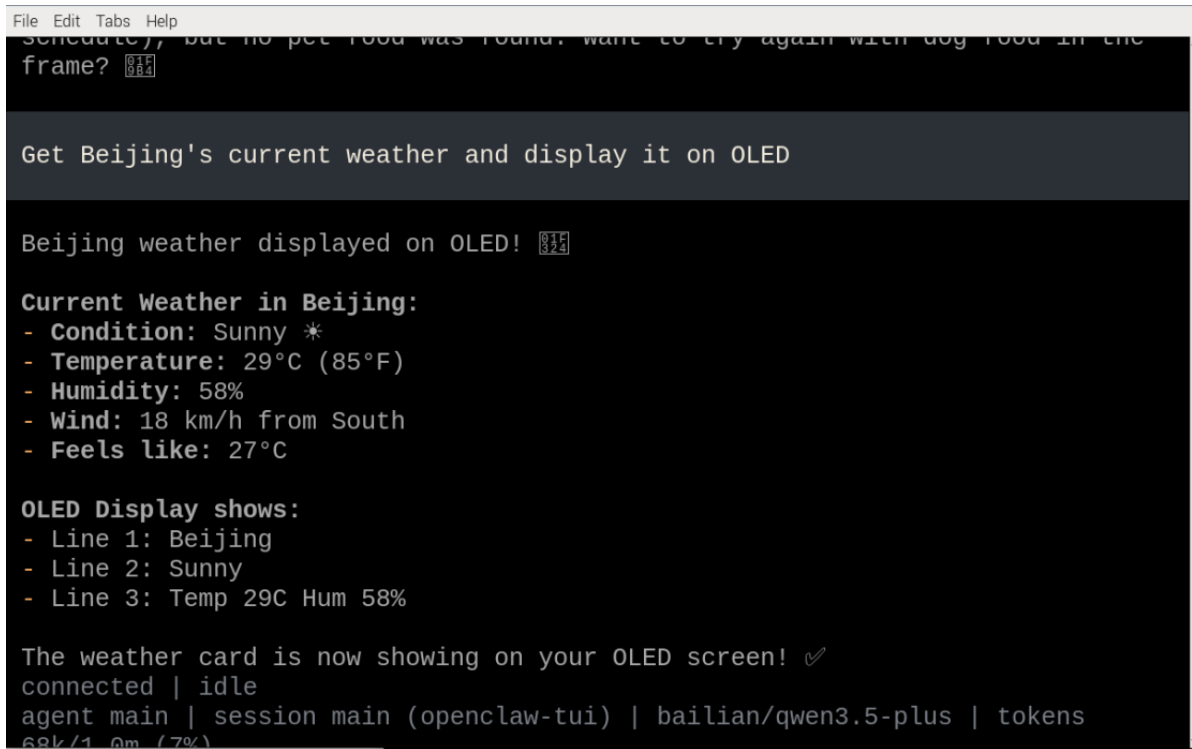
5.2 TUI Solution

If you have completed the TUI self-check in article 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Get current weather for Beijing and display it on OLED



```
File Edit Tabs Help
Schedule,, but no pet food was found. want to try again with dog food in the
frame? [0.1]

Get Beijing's current weather and display it on OLED

Beijing weather displayed on OLED! [0.1]

Current Weather in Beijing:
- Condition: Sunny *
- Temperature: 29°C (85°F)
- Humidity: 58%
- Wind: 18 km/h from South
- Feels like: 27°C

OLED Display shows:
- Line 1: Beijing
- Line 2: Sunny
- Line 3: Temp 29C Hum 58%

The weather card is now showing on your OLED screen! ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
68k/1.0m (7%)
```

5.3 Voice Interaction Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [04-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- AI weather station


```
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Ai weather station.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 📡 AI Weather Station Complete!

**Local Sensor Readings:**
-
-      **Indoor Temperature:** 30°C (86°F)
- **Indoor Humidity:** 51%
-      **Outdoor Weather (Shenzhen):**
- **Condition:** Partly cloudy 🌤️
-
-      **Outdoor Temperature:** 28°C (82°F)
- **Feels Like:** 35°C (95°F)
-
-      **Outdoor Humidity:** 84%
- **Wind:** 9 km/h from South
- **Rain
    Chance:** 72% today

**3-Day Forecast:**
- **Today (May 14):** High
    28°C, Low 23°C - Patchy rain nearby
- **Tomorrow (May 15):** High
    26°C, Low 23°C - Moderate rain expected
```

6. Project Implementation

Note: The following demonstrations are all conducted using `openclaw tui` dialogue. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Environment Collection

Experiment Objective

Read the local ambient temperature and humidity to complete the most basic data sampling for the weather station.

Effect Description

OpenClaw will call DHT to get the current environment data and return the results to the user, providing a real-time "weather reading" of the indoor environment.

Example Commands

- Read current ambient temperature and humidity
- What's the current indoor temperature and humidity
- Help me check the current weather data

6.2 Scenario 2: AI Broadcast

Experiment Objective

Make OpenClaw not just report numbers, but also summarize the environment status like a weather broadcaster.

Effect Description

OpenClaw can summarize "26 degrees, 57% humidity" into more natural phrases, such as "the environment is relatively comfortable", "it's a bit stuffy", "the air is slightly dry", etc.

Example Commands

- Give me a brief environment broadcast
- Is the environment comfortable now
- Make a weather summary based on current readings

6.3 Scenario 3: OLED Weather Card

Experiment Objective

Display local temperature and humidity results in a "weather card"-style page.

Effect Description

In addition to plain text results, OLED can also arrange location, status, temperature, and humidity into a compact card, making the entire project more like a desktop display device.

Example Commands

- Display an indoor weather card on OLED
- Get current weather for xx location and display it on OLED

6.4 Scenario 4: RGB Status Alert

Experiment Objective

Use RGB colors to quickly indicate the current environment status.

Effect Description

For example, turn on the green light when the environment is comfortable, yellow light when it's hot, red light when abnormal. This way, even without reading text, you can quickly know the current environment status.

Example Commands

- If the environment is normal, turn on green light
- If the temperature is high, turn on yellow light
- If abnormal, flash red light